System Design functional Decomposition
EchoKeys 49

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ECE 411: Industry Design Processes

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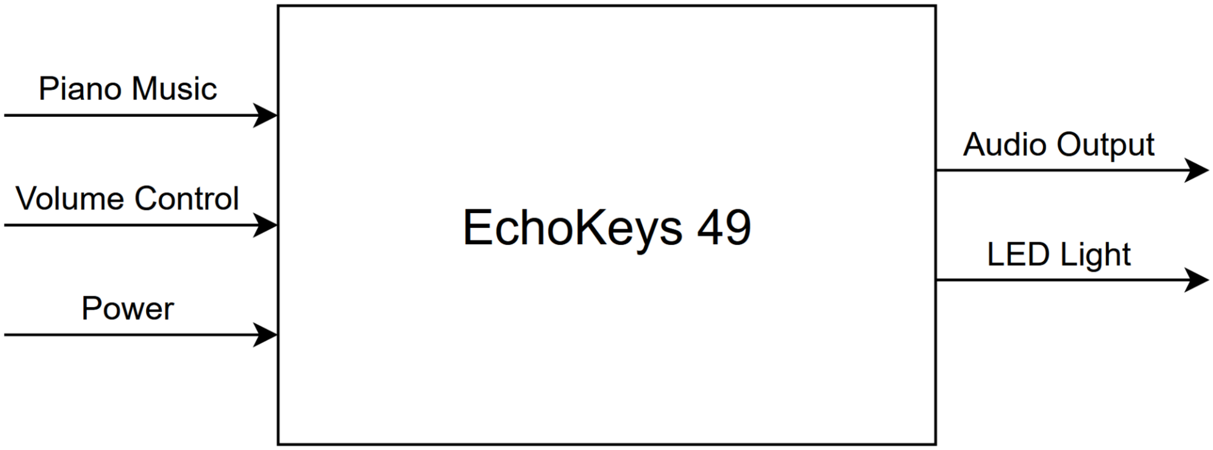
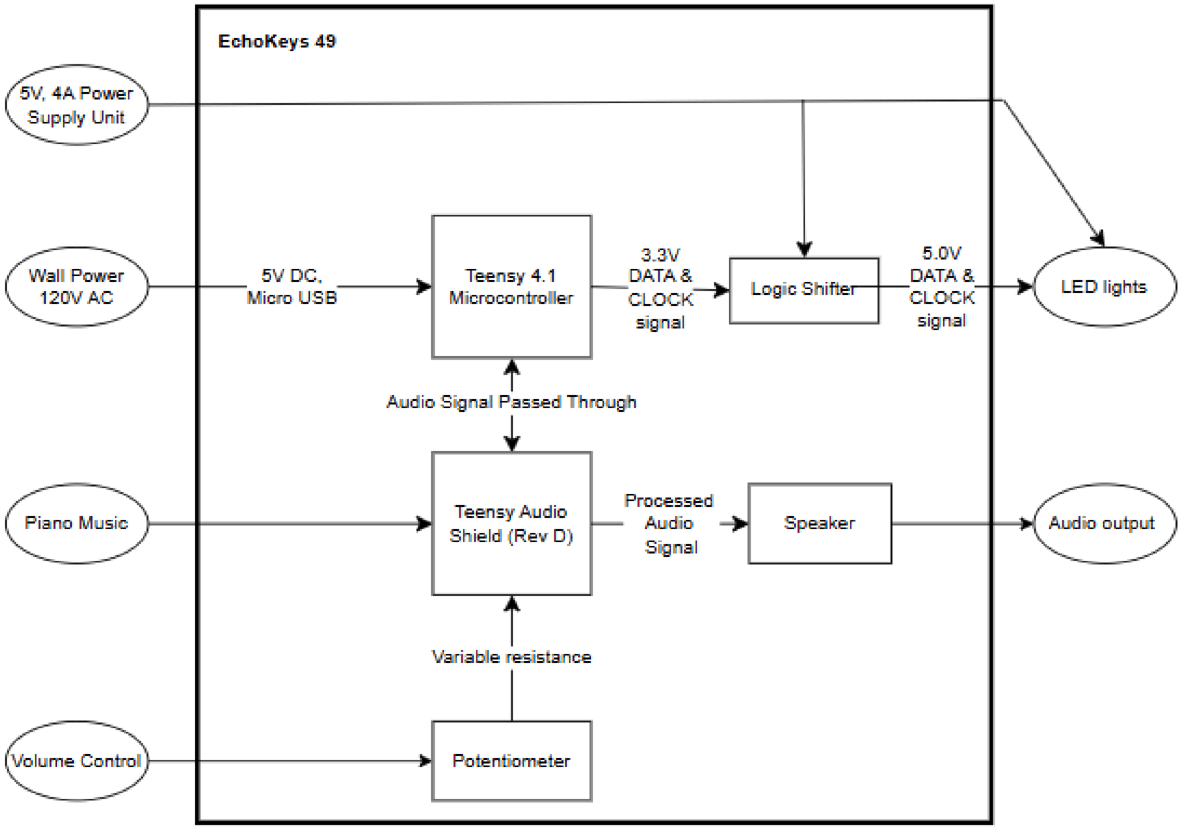
November 10th, 2025

Level 0:

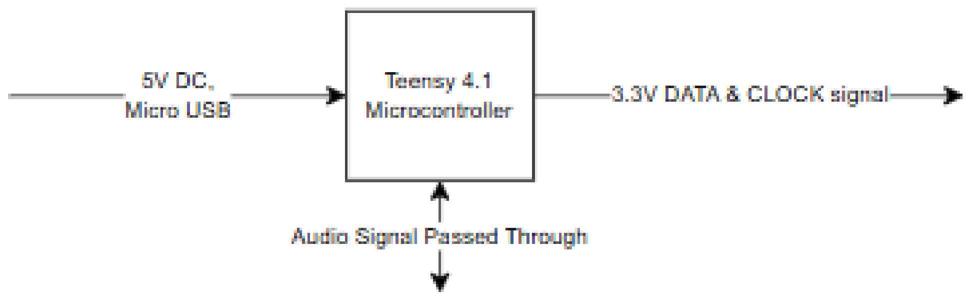


Module	EchoKeys 49 ~ "*49 keys, mirrored in light.*"
Inputs	• Piano Music: 0.5V peak • Volume Control: variable • Power: plugged into wall; 120V AC, 60Hz
Outputs	• Audio output • LED lights
Functionality	Take incoming piano music (500 mV max) and play the music through a speaker (variable) while showing which keys are being played.

Level 1:

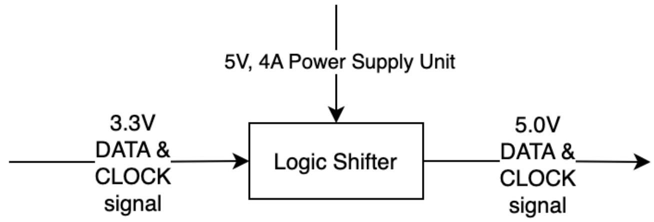

EchoKeys 49
5V, 4A Power Supply Unit
Wall Power 120V AC
5V DC, Micro USB
Teensy 4.1 Microcontroller
3.3V DATA & CLOCK signal
Logic Shifter
5.0V DATA & CLOCK signal
LED lights
Audio Signal Passed Through
Piano Music
Teensy Audio Shield (Rev D)
Processed Audio Signal
Speaker
Audio output
Variable resistance
Volume Control
Potentiometer

Teensy 4.1 Microcontroller



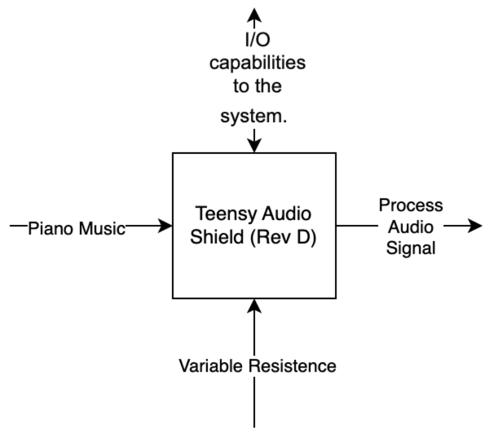
Modules	*Teensy 4.1 Microcontroller*
Inputs	5V DC Micro USB
Outputs	3.3V Data & Clock signal
Functionality	Controlling the appropriate LED to provide visual feedback on tuning.

Logic Shifter



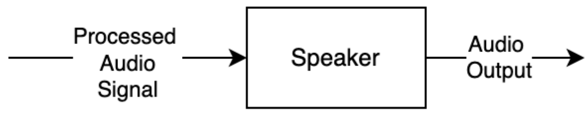
Modules	*Logic Shifter*
Inputs	3.3V Data & Clock signal, 5V/4A Power
Outputs	5.0V Data & Clock signal
Functionality	Safely translate the 3.3V logic signals from the teensy 4.1 to a higher voltage level.

Teensy Audio Shield (Rev D)



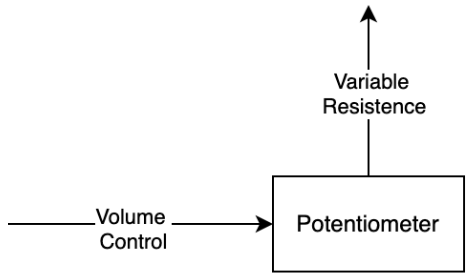
Modules	*Teensy Audio Shield (Rev D)*
Inputs	Piano Music, I/O Capabilities to the system, Variable resistance
Outputs	Process Audio Signal
Functionality	Acts as an ADC for passing digital audio to Teensy. Outputs passthrough audio to speaker, with pot for adjusting output volume.

Speaker



Modules	*Speaker*
Inputs	Processed Audio Signal
Outputs	Audio Output
Functionality	Provides audio passthrough of analog audio input

Potentiometer



Modules	*Potentiometer*
Inputs	User input
Outputs	Control voltage (for audio passthrough)
Functionality	Acts as a variable voltage divider, allowing the microcontroller to read position by converting the mechanical rotation of its knob into an analog voltage